

From Brazil: A Forest of Computer Graphics and Interactive Techniques

SIGGRAPH 2021 Retrospective Panel

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1 INTRODUCTION

This panel is devoted to the history of the deep, diverse, and rich contributions from Brazil to the global digital culture, from early Computer Art, the design of the Lua Programming Language to the use of digital media by contemporary indigenous communities. The panel has three presentations by key technologists, scholars, and curators:

1.1 “Waldemar Cordeiro and Giorgio Moscati: Pioneers of Computer Art in Brazil” by Caroline Menezes

This talk aims to present a brief overview of Italian-Brazilian artist Waldemar Cordeiro (1925-1973) with the emergence of pioneering experiences in computer art in Brazil. Together with the Italo-Brazilian physicist Giorgio Moscati (1934-), responsible for the programming of the algorithms, Cordeiro began, in 1968, to explore the use of computer resources to create a type of art in tune with the core of an industrial society that was already moving towards a computerized reality. Thus, the duo began to develop a series of works that aimed not to duplicate experiences already made in art but to expand them towards something unknown and, to a great extent, linked to the random aspect made possible by computer algorithms. Cordeiro and Moscati began to research the aesthetic possibilities of computers and seek to give legibility to the processes that take place within and around them. The loss of referential images, caused by the various abstract movements in art, pointed

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Cordeiro in a new direction, even more abstract, where art explains itself. These are the duo's ideals in their contact with computers, seeking a new humanism in the emerging computerized society. In conclusion, I aim to highlight the artistic legacy of the duo and how the brief period in which they worked together opened new horizons for artistic practice in the Brazilian context.

1.2 “The Lua Bar: Lua programming language” by Roberto Ierusalimsky

The Lua programming language was created in 1993 at the Pontifical Catholic University of Rio de Janeiro as an in-house tool for a project for the Brazilian Petroleum Corporation. Since then, it spread to become one of the leading scripting languages in the game industry. In this short presentation, I will discuss the path that Lua followed from that “unlikely place” for the birth of a programming language into the development teams of many world-renowned companies.

1.3 “Memory Practices in Indigenous Contemporary Cultures” by Louise Hisayasu

Memory Practices in Indigenous Contemporary Cultures: My research studies the impact that digital technologies and the Internet have had, as tools, on the safeguarding and endurance of first-person Indigenous memories. The inclusion of Indigenous communities in Brazil to the digital landscape has transformed remarkably since the early 2000's, when early digital inclusion programmes, called Pontos de Cultura Indígena (PCI) emerged in rural Indigenous villages. Since then, projects like Rádio Yandê, Índios Online and Arte Eletrônica Indígena have served as great case studies for Indigenous protagonism online. By reclaiming through indigenising, cultural producers have created new methods of communication, renewed aesthetics for Indigenous contemporary art and created novel tactics for sustainable ecological preservation. Pitman uses the term digital indigeneity to describe Indigenous peoples relationships to digital media, understood through their ways of appropriating new technologies and how they feel this relates to their greater sense of identity. Artist Denilson Baniwa has described his artistic practice as a rescue of Indigenous memory through the invasion or occupation of space. Spaces need to shift to *lugares de fala* (spaces for discussions) where cultural, ethnic and historical pluralities can exist in harmony, and where more visibility is given to voices which traditionally lie outside the hegemony of power.

2 BIOGRAPHIES

Dr Caroline Menezes. Dr Caroline Menezes is a curator, researcher, art critic and historian. She holds a PhD in Art History and Theory (University of the Arts London) and a master's degree in History of Art (University of Sussex). Menezes lives between Brazil and Germany and has curated exhibitions in England, Spain, Portugal and Brazil, where she curated exhibitions in institutions such as the Museum of Contemporary Art of Niterói, National Museum of the Republic, Brasília, and Oi Futuro, Rio de Janeiro. In the latter, the highlights were "Primary Codes," in 2015 an exhibition with around 100 artworks by computer artists Frieder Nake, Ernest Edmonds, Harold Cohen and Paul Brown, and the 2016 "More Performance Festival," featuring the show "Peter Weibel: The Messenger," dedicated to the work of Peter Weibel as a pioneer of public performance and its exchange with New Media Art. The More Performance festival was one of the outcomes of the awarded project "Virtual Bridges: Art and Technology between Brazil and Germany," which she developed in 2015-2016 at the ZKM | Zentrum für Kunst und Medien, Karlsruhe with the support of the Alexander von Humboldt Foundation. She is currently preparing a retrospective of the artist Tuneu at CCBB - Bank of Brazil Cultural Centre, in São Paulo. As an art critic, she worked for Studio International for over ten years. She is the author and editor of "Mais performance" and "primordiais" (Caosmos Publishers) and co-editor of *The Permanence of the Transient: Precariousness in Art* (Cambridge Scholars Publishing), among other books.

Roberto Ierusalimshchy. Roberto Ierusalimshchy is a Brazilian computer scientist, known for creating the Lua programming language. He holds a PhD in Computer Science from the Pontifical Catholic University of Rio de Janeiro where he has an appointment as an associate professor of informatics. He did a post-doc at University of Waterloo in 1992 and was visiting professor at Stanford University in 2012. He is the leading architect and the author of Programming

in Lua. He also created LPeg, a Lua library for implementing parsing expression grammars.

Louise M. Hisayasu (she/her). Louise M. Hisayasu (she/her) is an independent researcher and project coordinator based in Berlin, Germany. Currently, her work on The Glass Room is centred around promoting critical media literacy through art, creativity and education. Her postgraduate research drew connections between collective memory, digital technologies and Indigenous protagonism in Brazil, studying the movement through the lens of technology, media and culture production. The research mapped early digital inclusion and digital literacy programmes, as well as Indigenous-led art and digital culture projects to understand Indigenous peoples dynamic relationship to digital culture production and technology more broadly. Her work is primarily interested in understanding memory as being both individual and collective, simultaneously intangible and immortalized in things, and as being woven inter-generationally. Her personal research practice, Arquivo Tadaima which speculatively archives and collects objects and stories from the Japanese-Brazilian diaspora puts her own (hi)story and heritage in focus.

Moderator – May-Li Khoe. May-Li Khoe is an interdisciplinary artist and designer who is passionate about helping people cultivate their own creativity, curiosity, and critical thinking. When she's not busy building and empowering design teams, she's juggling one of too many side projects, usually involving some strategic combination of cultural practice, bright colors, faces-on-things, and glitter. In the past, May-Li invented new interactions across Apple's hardware and software platforms, built educational games at Leapfrog, and researched at the MIT Media Lab. Until recently, May-Li served as VP of Design at Khan Academy, developing a critically conscious practice of inclusiveness. She's currently taking time to recharge, write, and work on side projects. She also DJs, dances, and daydreams about more joyful ways to subvert the status quo.